

# Divya Mylavarpu

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## Professional Profile

Third-year Cyber Security Engineering student with practical experience in secure software development, vulnerability assessment, cryptography, and Linux-based security testing. Skilled in C/C++, Java, Python, Rust, and security tools including Burp Suite. Experienced in developing security-focused projects involving web application testing, authentication systems, and applied cryptography. Strong leadership and communication skills demonstrated through university leadership and mentoring roles. Seeking cybersecurity internship opportunities in security engineering, penetration testing, and secure software development.

## Technical Skills

**Programming:** C, C++, Java, Python, SQL, Rust, Bash, JavaScript, HTML, CSS

**Cyber Security:** OWASP Top 10, Vulnerability Assessment, Web App Security Testing, JWT, Linux Security, Network Security, Secure Coding, Authentication Systems

**Cryptography:** AES, ECB, CBC, CTR, GCM, PBKDF2, Cryptographic Hashing, Key Management

**Security Concepts:** Access Control, Authentication, Authorisation, Threat Modelling, Secure Development Lifecycle

**Tools:** Git, GitHub, Linux, Burp Suite, VS Code, IntelliJ IDEA, React, Vercel

**Software Engineering:** OOP, Data Structures & Algorithms, FSMs, Testing, Debugging, System Design

## Projects and Technical Work

### Automated Emergency Braking System (AEBS)

- Developed software components in Java and Rust for an automotive safety-critical system.
- Contributed to system testing, requirements analysis, documentation, and software design.
- Applied ISO 26262 principles including requirements traceability, testing documentation, and hazard analysis (HARA).
- Collaborated within a team environment using version control and structured development processes.

### Autonomous Vehicle Challenge (AVC) Robot

- Developed autonomous navigation and decision-making logic in C++.
- Integrated real-time sensor data with hardware control systems.
- Tested and refined robot performance under competition constraints.
- Applied debugging and optimisation techniques to improve reliability.

### Security & Web Application Testing

- Performed web application security testing using Burp Suite in Linux environments.
- Analysed web applications developed with SQL, Java, and Python technologies.
- Identified vulnerabilities including XSS, CSRF, SQL Injection, authentication flaws, and access control weaknesses.
- Applied secure coding and vulnerability mitigation principles based on industry best practices.

### Network Security & Authentication Systems

- Studied network security concepts including firewall design, network segmentation, and Zero Trust architectures.
- Analysed authentication mechanisms including session management, MFA, and JSON Web Tokens (JWTs).
- Investigated security controls and common attack vectors affecting modern web applications.

### Cybersecurity Portfolio Website

- Designed and developed a personal cybersecurity portfolio website using React and Vite.
- Implemented a Linux terminal inspired interface to showcase cybersecurity projects, technical skills, and professional experience.
- Built reusable React components with responsive layouts and modular CSS styling.

- Integrated external links, downloadable CV functionality, and project documentation to improve accessibility and user experience.
- Managed source control and project development using Git and GitHub.

## Applications of Cryptography

- Implemented AES encryption/decryption in Java across ECB, CBC, CTR, and GCM.
- Built CBC padding-oracle, ECB detection, weak-PIN brute-force, and CTR nonce-reuse attacks.
- Applied PBKDF2 key derivation, authenticated encryption, and secure key management.
- Tested cryptographic code with Gradle/JUnit while handling binary data and JCA APIs.

## Interactive Simulations and Games

- Created interactive simulations and games using Processing (Java).
- Implemented physics systems, collision detection, animation, and user interaction.
- Developed platformer and side-scrolling game mechanics.

## Cyber Security Experience

- Conducted web application security testing using Burp Suite, identifying vulnerabilities including XSS, CSRF, SQL injection, authentication flaws, and access control issues within OWASP top 10.
- Implemented and analysed cryptographic systems including AES encryption modes, padding oracle attacks, ECB detection, and CTR nonce reuse vulnerabilities.
- Practised Linux security concepts including privilege escalation, SetUID exploitation, vulnerability analysis and format string attacks.

## Leadership & Volunteering

### President — Women in Tech Club (University)

- Lead a student organisation encouraging women into technology fields.
- Organised networking and outreach events connecting students with industry professionals.
- Coordinate with university staff and industry speakers to facilitate events.

### UNI101 Mentor

- Mentored first-year students to support their transition into university life.
- Facilitated group sessions and provided academic and personal guidance.

## Work Experience

### Sales Assistant — Number One Shoes *Nov 2022 – Present*

- Delivered customer service in fast-paced retail environments.
- Developed communication, teamwork, and problem-solving skills.

### Tutor — Kumon *Jan 2023 – Dec 2023*

- Tutored students in mathematics and reading, improving their problem-solving and learning skills.
- Developed patience, communication, and the ability to explain complex ideas simply.

### After School Care Supervisor — YMCA *Jan 2023 – Apr 2024*

- Supervised and managed groups of children in a structured environment.
- Responsible for safety, organisation, and activity coordination.

## Education

### Bachelor of Engineering (Honours) — Cyber Security Engineering *Expected 2026/2027*

Victoria University, New Zealand